

# NanoGT Match Report: Leber hereditary optic neuropathy

**Disease:** Leber hereditary optic neuropathy (ORPHA:104)

**Primary gene:** MT-ND4

**Gene CDS:** 1377 bp

**Inheritance:** Mitochondrial inheritance

**Target tissues scored:** retina, CNS

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## Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.; Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints; Vector does not naturally cover all annotated disease tissues: cns, retina.

## Disease Mechanism Evidence

**Molecular mechanism:** mitochondrial loss of function

**Mechanistic detail:** Mitochondrial complex I dysfunction from MT-ND4 variant

**Gene-addition compatibility:** uncertain

**Preferred modality class:** specialized allotopic expression or mitochondrial strategy

**Evidence level/status:** direct / needs manual review

**Evidence summary:** Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific

**Evidence source:** [OMIM MT-ND4 gene entry](#)

## Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
  - Modality coverage is limited mainly to AAV and integrating ex vivo HSC vector precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
  - Endpoint readiness: retinal diseases often have measurable endpoints such as OCT, ERG, visual acuity, visual fields, or mobility testing, but genotype-specific progression still needs confirmation.
  - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
  - Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.
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## Top 5 GT Precedent Matches

| Rank | Program   | Vector | Score  | Confidence | Approval |
|------|-----------|--------|--------|------------|----------|
| 1    | CPCB-RPE1 | AAV8   | 6.4/10 | Medium     | phase2/3 |
| 2    | Skysona   | LV     | 6.3/10 | Medium     | approved |
| 3    | Luxturna  | AAV2   | 6.1/10 | Medium     | approved |
| 4    | Libmeldy  | LV     | 5.9/10 | Medium     | approved |
| 5    | Hemgenix  | AAV5   | 5.6/10 | Medium     | approved |

## Match #1: CPCB-RPE1

**Precedent disease:** Achromatopsia

**Vector:** AAV8

**Tissue target:** retina/photoreceptor

**Composite score:** 6.4 / 10

### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Packaging fit             | 2.00        | 2.0         | Gene CDS size vs vector cargo capacity                       |
| Tissue tropism            | 1.50        | 2.0         | Vector naturally reaches disease target tissue               |
| Protein class             | 1.50        | 2.0         | Same se-creted/lysosomal/membrane/intracellular class        |
| Pathway similarity        | 2.00        | 2.0         | Same or related biological pathway                           |
| Modality compatibility    | 1.00        | 2.0         | Disease mechanism supports gene-addition precedent           |
| Inheritance compatibility | 0.30        | 1.0         | AR/XL loss-of-function pattern match                         |
| Approval precedent        | 0.70        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 1.00        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 1.00        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 0.80        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 0.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>6.43</b> | <b>10.0</b> | Raw sum / 21 × 10 (v2: 14 dimensions)                        |

### Rationale

- Gene CDS 1377bp / cargo 4700bp (29% utilized)
- Precedent target match: retina
- Both membrane proteins

- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Pathway match: retinal\_phototransduction
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: phase2/3
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- $\beta$ 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene. Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

## Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, retina
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

## Match #2: Skysona

**Precedent disease:** Cerebral adrenoleukodystrophy

**Vector:** LV

**Tissue target:** hematopoietic/CNS

**Composite score:** 6.3 / 10

### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Packaging fit             | 2.00        | 2.0         | Gene CDS size vs vector cargo capacity                       |
| Tissue tropism            | 1.50        | 2.0         | Vector naturally reaches disease target tissue               |
| Protein class             | 1.50        | 2.0         | Same secreted/lysosomal/membrane/intracellular class         |
| Pathway similarity        | 0.50        | 2.0         | Same or related biological pathway                           |
| Modality compatibility    | 1.00        | 2.0         | Disease mechanism supports gene-addition precedent           |
| Inheritance compatibility | 0.30        | 1.0         | AR/XL loss-of-function pattern match                         |
| Approval precedent        | 1.00        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 2.00        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 1.00        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 0.80        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 0.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>6.33</b> | <b>10.0</b> | Raw sum / $21 \times 10$ (v2: 14 dimensions)                 |

### Rationale

- Gene CDS 1377bp / cargo 8000bp (17% utilized)
- Precedent target match: cns
- Both membrane proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (retinal phototransduction vs peroxisomal)
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allo-topic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden

- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- $\beta$ 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene. Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

### Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, retina
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities

### Match #3: Luxturna

**Precedent disease:** Leber congenital amaurosis type 2

**Vector:** AAV2

**Tissue target:** retina/RPE

**Composite score:** 6.1 / 10

### Score Breakdown

| Dimension      | Score | Max | What it measures                               |
|----------------|-------|-----|------------------------------------------------|
| Packaging fit  | 2.00  | 2.0 | Gene CDS size vs vector cargo capacity         |
| Tissue tropism | 2.00  | 2.0 | Vector naturally reaches disease target tissue |

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Protein class             | 0.50        | 2.0         | Same se-creted/lysosomal/membrane/intracellular class        |
| Pathway similarity        | 2.00        | 2.0         | Same or related biological pathway                           |
| Modality compatibility    | 1.00        | 2.0         | Disease mechanism supports gene-addition precedent           |
| Inheritance compatibility | 0.30        | 1.0         | AR/XL loss-of-function pattern match                         |
| Approval precedent        | 1.00        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 0.50        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 1.00        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 0.80        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 0.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>6.10</b> | <b>10.0</b> | Raw sum / $21 \times 10$ (v2: 14 dimensions)                 |

## Rationale

- Gene CDS 1377bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: retina
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Pathway match: retinal\_visual\_cycle
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allo-topic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: approved
- Vector immunogenicity (AAV2): very high (~55%) — majority of patients may be ineligible; major trial design challenge
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- $\beta$ 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene.

Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

## Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

## Match #4: Libmeldy

**Precedent disease:** Metachromatic leukodystrophy

**Vector:** LV

**Tissue target:** hematopoietic/CNS

**Composite score:** 5.9 / 10

## Score Breakdown

| Dimension                 | Score | Max | What it measures                                     |
|---------------------------|-------|-----|------------------------------------------------------|
| Packaging fit             | 2.00  | 2.0 | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 1.50  | 2.0 | Vector naturally reaches disease target tissue       |
| Protein class             | 0.50  | 2.0 | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 0.50  | 2.0 | Same or related biological pathway                   |
| Modality compatibility    | 1.00  | 2.0 | Disease mechanism supports gene-addition precedent   |
| Inheritance compatibility | 0.30  | 1.0 | AR/XL loss-of-function pattern match                 |
| Approval precedent        | 1.00  | 1.0 | Regulatory approval / trial stage                    |
| Immunogenicity            | 2.00  | 2.0 | Pre-existing NAb seroprevalence for this vector      |

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 1.00        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 0.80        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 0.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>5.86</b> | <b>10.0</b> | Raw sum / $21 \times 10$ (v2: 14 dimensions)                 |

## Rationale

- Gene CDS 1377bp / cargo 8000bp (17% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (retinal phototransduction vs leukodystrophy)
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- $\beta$ 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene. Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

## Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear



AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, retina
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities

## Match #5: Hemgenix

**Precedent disease:** Hemophilia B

**Vector:** AAV5

**Tissue target:** liver

**Composite score:** 5.6 / 10

### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Packaging fit             | 2.00        | 2.0         | Gene CDS size vs vector cargo capacity                       |
| Tissue tropism            | 1.00        | 2.0         | Vector naturally reaches disease target tissue               |
| Protein class             | 0.50        | 2.0         | Same secreted/lysosomal/membrane/intracellular class         |
| Pathway similarity        | 0.50        | 2.0         | Same or related biological pathway                           |
| Modality compatibility    | 1.00        | 2.0         | Disease mechanism supports gene-addition precedent           |
| Inheritance compatibility | 0.30        | 1.0         | AR/XL loss-of-function pattern match                         |
| Approval precedent        | 1.00        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 2.00        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 0.20        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 1.00        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 0.80        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 0.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>5.62</b> | <b>10.0</b> | Raw sum / 21 × 10 (v2: 14 dimensions)                        |

## Rationale

- Gene CDS 1377bp / cargo 4700bp (29% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (retinal\_phototransduction vs coagulation)
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- $\beta$ 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene. Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

## Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: retina
- Only partial tissue match; verify target-cell transduction and delivery route manually
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone